

Notes: Célérier & Vallée (QJE, 2017)

Catering to Investors Through Security Design: Headline Rate and Complexity

Contents

1	Big picture	3
1.1	Incremental contribution of the paper	3
2	Data and measurement	4
2.1	Retail structured product market	4
2.2	Sample construction	4
2.3	Headline rate	4
2.4	Complexity measures	5
2.5	Risk: complete loss exposure	5
2.6	Markup and ex post performance	5
2.7	Bank and distributor characteristics	5
3	Empirical specifications and conceptual framework	6
3.1	Conceptual framework: risk sharing versus catering	6
3.2	Headline rate and complexity	6
3.3	Complexity and product risk	6
3.4	Low-interest-rate environments	6
3.5	Distributor and bank-characteristic regressions	7
3.6	Profitability regressions	7
4	Identification and empirical strategy	7
4.1	Nature of identification	7
4.2	Text-based measurement strategy	7
4.3	Product-level controls and fixed effects	8
4.4	Interest-rate variation	8
4.5	Distributor heterogeneity	8
4.6	Profitability evidence	8
4.7	Threats and limitations	9
5	Tables and figures	9
5.1	Figure I: Volumes Sold per Year and Figure II: Example of a Retail Structured Product Payoff	9
5.2	Table I: Retail Structured Product Names in France: Key Analogies and Table II: Measuring Complexity	10
5.3	Table III: Summary Statistics	11
5.4	Figure III: Headline Rate and Benchmark Interest Rate and Figure IV: Evolution of Product Complexity	11

5.5	Figure V: Share of Volumes Exposing Investors to Total Losses and Figure VI: Headline Rate by Complexity Levels	12
5.6	Table IV: Headline Rate, Product Complexity and Risk	13
5.7	Figure VII: Share of Volumes Exposing Investors to Complete Losses by Complexity Levels and Table V: Complete Loss Exposure and Product Complexity	14
5.8	Table VI: Headline Rate, Complexity, and Risk in Low Interest Rate Environments	14
5.9	Table VII: Complexity Measures and Distributor Characteristics	15
5.10	Table VIII: Headline Rates, Complexity, and Profitability	15
6	Interpretive synthesis	16
7	Short summary	16
8	Conclusion	17
8.1	Core conclusion	17
8.2	Economic intuition	17
8.3	Incremental contribution revisited	17
8.4	Policy implication	17
8.5	Takeaway	17

1 Big picture

- **Question.** How do financial institutions design retail financial products when households seek high advertised yields but may underweight hidden risks and payoff complexity?
- **Market.** The paper studies European retail structured products from 2002 to 2010, a large market in which banks design payoff formulas and sell products directly to households.
- **Key product feature.** Retail structured products often advertise a high *headline rate*: the return in the best-case scenario. This rate is salient in marketing but is embedded in a payoff formula with downside conditions, barriers, autocall clauses, reverse convertibility, and other features.
- **Core empirical fact.** As benchmark interest rates fall, headline rates do not fall one-for-one. Instead, banks keep headline rates high by increasing payoff complexity and investor exposure to complete losses.
- **Headline rate and complexity.** Products with more payoff features, more scenarios, and longer payoff descriptions offer higher headline rates.
- **Risk.** More complex products are more likely to expose investors to complete loss of principal.
- **Bank heterogeneity.** Savings banks, private banking institutions, and risk-taking banks distribute more complex and riskier products than commercial banks.
- **Profitability.** More complex, higher-headline-rate, and riskier products have higher markups for banks and lower ex post performance for investors.
- **Main interpretation.** Complexity is not only a tool for risk sharing. It is also a way to cater to household demand for salient yield while shrouding risks.
- **Takeaway.** Security design in retail finance can be shaped by investor psychology: banks engineer complexity to preserve attractive headline rates when low interest rates make simple products unattractive.

1.1 Incremental contribution of the paper

- **New data contribution.** The paper builds a comprehensive data set of roughly 55,000 retail structured products issued in 16 European countries, with detailed payoff descriptions, distributors, countries, maturities, volumes, underlying assets, and product types.
- **Measurement contribution.** It converts unstructured payoff descriptions into quantitative measures of headline rate, complexity, and risk through text analysis. This is important because product complexity is otherwise hard to compare across thousands of heterogeneous payoff formulas.
- **Behavioral finance contribution.** It connects security design to investors' tendency to focus on salient high yields. The paper provides product-level evidence that banks may cater to yield-seeking behavior by embedding high headline rates inside complex structures.
- **Security design contribution.** Standard theories emphasize innovation for risk sharing. This paper shows that complexity can also be a by-product of marketing and shrouding. The same derivative technology that allows efficient risk sharing can be used to obscure risk.
- **Household finance contribution.** The paper links product design, bank type, and investor sophistication: banks serving less sophisticated clients distribute more complex products.
- **Banking contribution.** The paper shows that risk-taking banks offer more complex and riskier products, connecting retail product design to bank balance sheet incentives and profitability.
- **Welfare contribution.** By documenting higher markups and worse ex post performance for more complex products, the paper provides evidence that product complexity can redistribute surplus from households to financial intermediaries.

2 Data and measurement

2.1 Retail structured product market

- Retail structured products are investment products sold to households whose payoff is defined ex ante as a nonlinear function of an underlying asset.
- Underlying assets include equity indexes, single stocks, interest rates, exchange rates, commodities, and other indexes.
- The products are built using embedded derivatives, which gives banks large flexibility in designing payoff formulas.
- The European market is especially useful because it is large, active, and covered by detailed product-level data.

Interpretation: The setting is ideal for studying endogenous security design because the bank chooses the payoff formula, the formula is observable, and the product is marketed to retail investors.

2.2 Sample construction

- The main data cover core retail structured products issued in Europe from 2002 to 2010.
- The sample covers 16 countries and more than 400 distributors.
- Product-level information includes issuance volume, maturity, country, distributor, product format, underlying asset class, and a detailed English description of the payoff formula.
- The full market sample contains 53,541 products for complexity and risk analysis.
- The headline-rate analysis focuses on coupon products because headline rates are clearly defined for products promising a fixed rate in the best scenario.
- The markup analysis uses a smaller pricing sample of Euro Stoxx 50 products, for which the authors can calculate fair values using option-pricing methods.
- The ex post performance analysis uses products that matured before 2010.

Interpretation: The paper combines breadth and depth: the broad sample identifies market-wide design patterns, while the smaller pricing sample measures whether complexity is profitable for banks.

2.3 Headline rate

- The headline rate is the fixed yearly return the investor receives in the best possible scenario.
- The paper identifies headline rates through text analysis of product payoff descriptions.
- The main headline-rate outcome is the spread between the product headline rate and the benchmark interest rate, usually the five-year swap rate at issuance.

Interpretation: The headline rate is the product attribute most likely to be salient to yield-seeking investors. The spread to benchmark rates measures how much the advertised return departs from the prevailing interest-rate environment.

2.4 Complexity measures

The paper develops three main measures of payoff complexity:

- **Number of features.** A text-analysis algorithm identifies payoff features such as digital coupons, knockouts, barriers, reverse convertibles, autocalls, or capital protection clauses.
- **Number of scenarios.** The authors count the number of conditional payoff scenarios in the payoff description.
- **Description length.** The number of characters in the payoff formula measures how long and difficult the formula is to describe.

Interpretation: These measures capture complementary dimensions. Features capture contractual building blocks; scenarios capture conditional branching; length captures the overall burden of reading and comparing the contract.

2.5 Risk: complete loss exposure

- The main risk indicator equals one if the investor can lose the entire initial investment.
- The indicator is obtained from payoff descriptions and identifies products whose minimum final payoff can be zero.
- This risk measure focuses on extreme downside risk rather than variance or expected return.

Interpretation: Complete loss exposure is a simple but powerful risk indicator for retail investors. It captures whether principal protection is absent in the worst state.

2.6 Markup and ex post performance

- Markup is the difference between the product issue price and the fair value at issuance.
- Fair values are calculated with option-pricing methods, including Longstaff-Schwartz Monte Carlo and local volatility for products indexed to the Euro Stoxx 50.
- Markups are annualized for comparability across maturities.
- Ex post performance is the realized yearly return of products that matured before 2010.

Interpretation: Markup measures the issuer/distributor's ex ante revenue from product design. Ex post performance asks whether products that are complex and profitable for banks also deliver worse outcomes to investors.

2.7 Bank and distributor characteristics

- Distributors are classified as commercial banks, savings banks, private banks/wealth managers, insurance firms, and other intermediaries.
- Savings banks tend to target rural and lower- to middle-income households.
- Bank risk-taking proxies include leverage, reliance on wholesale funding, and exposure to Greek sovereign debt.
- Bank balance sheet data are linked to product distributors.

Interpretation: Distributor heterogeneity allows the paper to connect product design to client sophistication and bank risk-taking incentives.

3 Empirical specifications and conceptual framework

3.1 Conceptual framework: risk sharing versus catering

The paper contrasts two theories:

- **Risk-sharing view.** Complexity expands the set of payoffs and can improve investor risk sharing.
- **Catering/shrouding view.** Complexity helps banks offer salient high headline rates while making downside risk less transparent.

Interpretation: Both theories can explain why complex securities exist. The identifying content comes from patterns that are difficult for pure risk-sharing theories: complexity rises when interest rates are low, complexity is more profitable to banks, and less sophisticated client bases receive more complex products.

3.2 Headline rate and complexity

The main cross-sectional regression is:

$$HeadlineRate_i = \alpha Complexity_i + \beta X_i + \delta_y + \theta_c + \eta_d + \varepsilon_i. \quad (1)$$

- $HeadlineRate_i$ is the spread between product headline rate and benchmark interest rate.
- $Complexity_i$ is alternatively number of features, number of scenarios, or description length.
- X_i includes product characteristics such as underlying asset, format, maturity, and volume.
- δ_y , θ_c , and η_d are year, country, and distributor fixed effects.

Interpretation: A positive coefficient means that products with more complex payoff formulas advertise higher returns, holding constant the observable product category, issuer/distributor, country, and year.

3.3 Complexity and product risk

The paper estimates logit regressions of complete loss exposure on complexity:

$$Pr(CompleteLoss_i = 1) = \Lambda(\alpha Complexity_i + \beta X_i + \delta_y + \theta_c + \eta_d). \quad (2)$$

Interpretation: The risk regression asks whether complexity is merely cosmetic or whether it is associated with materially worse downside states for investors.

3.4 Low-interest-rate environments

The interest-rate specification is:

$$Y_{i,c,t} = \alpha BenchmarkRate_{c,t} + \beta X_i + \delta_t + \theta_c + \eta_d + \varepsilon_{i,c,t}, \quad (3)$$

where $Y_{i,c,t}$ is headline-rate spread, product complexity, or complete loss exposure. Some specifications interact the benchmark rate with a country's pre-euro high-rate exposure.

Interpretation: This design asks whether banks respond to low interest rates by engineering products that maintain high salient returns. The interaction with pre-euro rate experience captures whether countries used to high yields respond more strongly.

3.5 Distributor and bank-characteristic regressions

The paper estimates regressions of headline rate, complexity, and complete loss exposure on bank type and risk-taking proxies:

$$Y_i = \alpha BankCharacteristic_d + \beta X_i + \delta_t + \eta_{product} + \varepsilon_i. \quad (4)$$

Interpretation: These specifications test whether product complexity is related to the seller’s client base and balance sheet risk appetite.

3.6 Profitability regressions

The profitability tests estimate:

$$Markup_i = \alpha Z_i + \beta X_i + FE + \varepsilon_i, \quad (5)$$

where Z_i is headline-rate spread, complexity, or complete loss exposure. The paper also estimates analogous regressions with ex post product return as the dependent variable.

Interpretation: If complexity is part of a shrouding/catering strategy, it should be associated with higher bank markups and poorer investor performance. If it only improved risk sharing, higher complexity would not necessarily predict higher bank surplus.

4 Identification and empirical strategy

4.1 Nature of identification

- The paper is not a randomized experiment. It is a product-level empirical design that combines text-based measurement, rich fixed effects, and cross-sectional/time-series variation.
- The main identification goal is to distinguish a pure risk-sharing story from a catering/shrouding story.
- The design relies on patterns across products, time, countries, distributors, and banks.

Interpretation: The paper’s claim is strongest when multiple margins point in the same direction: low rates, high headline rates, rising complexity, higher risk, higher bank markups, and worse investor performance.

4.2 Text-based measurement strategy

- The key measurement innovation is algorithmic extraction of headline rates, payoff features, scenario counts, and risk exposure from textual payoff formulas.
- This creates consistent variables across tens of thousands of heterogeneous products.
- The paper uses the same stable source of translated payoff descriptions, reducing concern that increases in complexity are merely disclosure changes.

Interpretation: Objective text analysis is central. Without it, the paper could not compare complexity across countries, issuers, and years.

4.3 Product-level controls and fixed effects

- Product controls include underlying asset class, product format, maturity, volume, and other design characteristics.
- Year fixed effects absorb market-wide trends.
- Country fixed effects absorb stable country differences.
- Distributor fixed effects compare products within the same distributor.
- Some specifications include quarter fixed effects or year-by-maturity fixed effects.

Interpretation: The fixed effects make the analysis more demanding. For example, the headline-rate/complexity relation is not only a comparison of different countries or different banks; it persists within controlled product environments.

4.4 Interest-rate variation

- The paper studies how product design changes when benchmark interest rates fall.
- The main variation comes from interest rates across countries and over time.
- The authors also use countries' pre-euro interest-rate histories as a proxy for investors' past exposure to high-yield environments.

Interpretation: Low interest rates are the demand shock to yield appetite. If investors anchor on high returns, banks can respond by raising complexity and risk to preserve high headline rates.

4.5 Distributor heterogeneity

- Savings banks are interpreted as serving a less sophisticated retail client base.
- Private banks and commercial banks serve different client segments.
- If less sophisticated households are more vulnerable to shrouding, product complexity should be higher in savings-bank distribution.

Interpretation: The distributor results help move from product design to household finance: complexity is not only a technological outcome, but also varies with the likely sophistication of the buyer pool.

4.6 Profitability evidence

- Markup regressions ask whether complex products are more profitable to banks at issuance.
- Ex post return regressions ask whether complexity is associated with lower realized returns for investors.
- The two outcomes are complementary: high markups are the bank-side payoff; low performance is the investor-side cost.

Interpretation: Profitability is a key discriminator between risk-sharing and shrouding. Products that simply improved risk sharing need not systematically have higher markups and worse performance.

4.7 Threats and limitations

- Complexity may proxy for unobserved investor demand or unobserved hedging costs.
- Markup estimates are available only for a subset of products.
- Ex post performance is affected by realized market outcomes and survivorship of matured products.
- The paper cannot observe investors' beliefs, attention, financial literacy, or suitability at the individual level.
- Bank type and bank risk-taking are not randomly assigned.

Interpretation: The evidence is best read as a coherent set of conditional correlations and mechanism tests, not as a single causal experiment. Its strength is the consistency of the patterns across product design, interest rates, bank type, markups, and performance.

5 Tables and figures

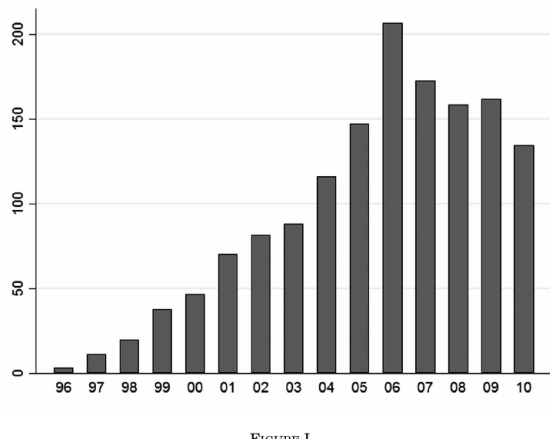
5.1 Figure I: Volumes Sold per Year and Figure II: Example of a Retail Structured Product Payoff

Read:

- Figure I shows rapid growth in issuance volumes from the late 1990s through 2006, followed by a smaller decline after the global financial crisis.
- Figure II illustrates the central marketing logic: the product advertises a high annual coupon in the favorable scenario, but the payoff formula contains conditions that expose the investor to losses if the underlying falls below thresholds.
- The headline rate is visible and simple; the full payoff is conditional and harder to process.

Interpretation: Figure I establishes that the market is economically important. Figure II shows why headline rate and complexity can coexist: a product can look attractive in the best case while embedding difficult-to-evaluate downside exposure.

Economic message: The basic fact motivating the paper is that a large household market grew around products whose salient promised yield is easier to understand than the risks needed to generate it.



(a) Figure I: Volumes sold per year

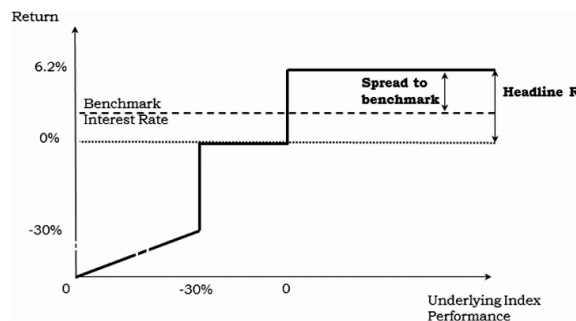


FIGURE II

Example of a Retail Structured Product Payoff

(b) Figure II: Example payoff

5.2 Table I: Retail Structured Product Names in France: Key Analogies and Table II: Measuring Complexity

Read:

- Table I shows that French product names use positive metaphors: transformation, balance, journey, and resource.
- These names transfer attributes such as vitality, security, adventure, privilege, and opportunity to the product.
- Table II shows how the authors’ algorithm translates payoff text into headline rate, number of features, scenarios, description length, and complete loss exposure.
- The Fixeo example has more features, more scenarios, and complete loss exposure, whereas the simpler example has fewer features and no complete loss exposure.

Interpretation: Table I documents the marketing layer: product names are designed to frame complex products positively. Table II documents the measurement layer: complex legal/payoff language is transformed into quantitative variables.

Economic message: The paper links marketing language and security design. Banks do not simply design payoffs; they also present them in a way that directs attention to upside attributes.

Key analogies (%)	Transferred attributes (%)	Examples
Transformation (32)	Vitality (11)	Dynamic, Elanceo, Energetic, Expansia
	Amplification (9)	Maximizer, Melioris, Optimiz, Digimax
	Success (6)	Winner, Best seller, Emeritus, Star
	Multiplication (6)	Double top, Triple horizon
Balance (25)	Security (18)	Guarantee, Amareo, Locker, Serenity
	Robustness (5)	Strength, Magnesium, Lion, Protein
	Stability (1)	Beau fixe
	Uncharted territories (6)	Archipel, Chamsin, Wapiti, Jayanne
Journey (24)	Adventure (4)	Conquistador, Drakkar, Cruzador
	Alpinism (2)	Cordillera, Hight, Hiking, Yeti
	Mythology (2)	Izeis, Goliath, Keops, Nemea
	Cap (1.4)	Objective, Cap, Horizon
Resource (19)	Exotic culinaryity (1.2)	Capuccino, Pimento, Lion, Cardamone
	Virtuosity (6)	Allegro, Arpeggio, Bolero, Harmony
	Privilege (5)	Four stars, Diamond, Quartz, Signature
	Magic (4)	Prism, Filtreo, Elixir, Hologram
	Opportunity (2)	Opportunity, Declic, Atout
	Sport (2)	Sprint, Tie Break, Triathlon
	Strategy (1.2)	Strategy, Selection, Allocator
	Precision (1)	Metronom, Autofocus, Zoom
	Science (1)	Alpha, Elipse, Isocel, Philosophy
	Innovation (0.7)	Digiteo, Primio, Inedit

Notes. This table provides the frequency of key analogies and transferred attributes used in the names of French retail structured products. The typology of analogies is from Zaltman (1997). The sample covers all products issued in France from 2002 to 2010.

(a) Table I: product names and analogies

	Example 1: Borsa Protetta Arancia, November 2009	Example 2: Fixeo
Details		
Year	2009	2010
Country	Italy	France
Provider	ING	Credit Agricole
Maturity	1	3
Description	This product is linked to the performance of the DJ Euro Stoxx 50. At maturity, [if the underlying index registers a level equal to or higher than] 70% of its strike level, [the product offers a capital return of 104% of the initial investment](1). Otherwise, the product offers a capital return of 70% the initial investment.	This is a growth product linked to the DJ Eurostoxx50. After 1.5 years of investment, [if] the level of the index is at or above its initial level, then [the product terminates](1) on that date and offers a capital return of 112% at that time. At maturity, the product [offers a capital return of 124%, as long as](2) the final index level is at or above its initial level. [Otherwise], the product offers a capital return of 100%, as long as the final index level is at or above 60% of its initial level. [In all other cases], the product offers a capital return of 100%, [decreased by the fall in the index](3) over the investment period.
Headline rate	4%	8%
Complexity measures		
# Features	1	3
# Scenarios	2	4
Length	308	636
Payoff features	(1) Digital	(1) Knockout (2) Digital (3) Reverse convertible
Complete loss exposure	No	Yes

Notes. (x): Text identifying payoff x. This table shows how our text analysis algorithm collects from two actual product descriptions the product headline rate, then quantitative measures of complexity: number

(b) Table II: measuring complexity

5.3 Table III: Summary Statistics

Read:

- The average headline coupon is 8.2 percent; the volume-weighted average is 7.4 percent.
- The average spread to benchmark is 4.6 percent, showing that headline rates are far above risk-free rates.
- The average product has 2.5 payoff features, 2.2 scenarios, and a payoff description length of 510 characters.
- 29 percent of products expose investors to complete loss.
- The average yearly markup is 0.76 percent, or 1.4 percent including disclosed fees.
- The average ex post yearly return is about 2 percent, with a wide range from large losses to large gains.

Interpretation: The summary statistics reveal that complexity, risk, and profitability are not rare outliers. They are central features of the retail structured product market.

Economic message: The typical product offers a high advertised yield but embeds multiple contingencies and meaningful downside risk.

	Average	Weighted average	Std. dev.	Min	p25	p75	Max	<i>N</i>
adline rate	8.2	7.4	3.7	1.0	5.2	10.0	25.0	26,352
spread to benchmark, in %	4.6	3.8	3.7	-3.8	2.0	6.50	22.6	26,352
complexity measures								
! Features	2.5	2.4	1	1	2	3	7	53,541
! Scenarios	2.2	2.1	1.5	1	1	3	16	53,541
length	510	519	207	140	361	630	2,124	53,541
is exposure								
indicator variable	0.29	0.16	—	—	—	—	—	53,541
markup								
product yearly markup, in %	0.76	0.90	1.3	-1.8	0.1	1.3	12.5	141
including disclosed fees	1.4	2.0	1.6	-1.4	0.3	1.9	12.5	141
post performance								
product yearly return, in %	2	2.2	7.7	-81.0	0.0	4.7	125.3	5,841
times								
n 2010 million euros	21	—	67	0.0	4.4	19.0	3,106.6	46,613

Notes: This table displays summary statistics for the three measures of complexity developed in the study. Number of features is obtained through a text analysis of the detailed off description, number of scenarios by counting the number of conditions in the product description, and length by counting the number of characters of the payoff description. Adline rate is defined for coupon products as the fixed rate that the investor receives in the best possible scenario. Complete Loss exposure is an indicator variable equal to 1 if the issuer is exposed to total losses. Markup is defined as the difference between insurance price and the fair value at issuance calculated using a local volatility diffusion.

Table III: Summary Statistics

5.4 Figure III: Headline Rate and Benchmark Interest Rate and Figure IV: Evolution of Product Complexity

Read:

- Figure III shows that the headline-rate spread rises when benchmark interest rates fall.
- The benchmark interest rate drops sharply after the financial crisis, but headline-rate spreads remain high.
- Figure IV shows increasing product complexity over time, measured by features, scenarios, and payoff-description length.
- Complexity continues rising with no clear crisis-period reversal.

Interpretation: These figures are central to the catering interpretation. When simple safe yields are low, banks appear to preserve attractive headline rates by changing the product structure rather than lowering advertised returns.

Economic message: Falling interest rates do not simply make retail products offer lower yields; they induce a redesign of products toward greater complexity.

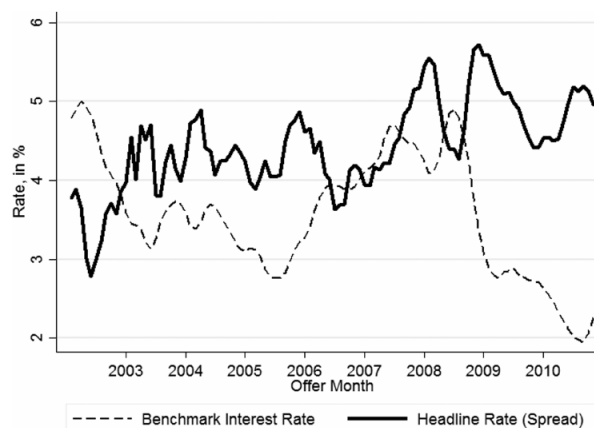


FIGURE III

(a) Figure III: headline rate and benchmark rate

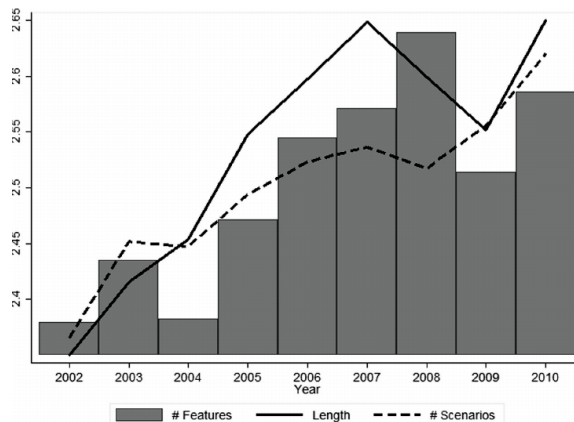


FIGURE IV

Evolution of Product Complexity

This figure shows the predicted complexity of retail structured products by year

(b) Figure IV: evolution of product complexity

5.5 Figure V: Share of Volumes Exposing Investors to Total Losses and Figure VI: Headline Rate by Complexity Levels

Read:

- Figure V shows that the share of issuance volume exposing investors to complete losses increases strongly over the sample.
- Figure VI shows that higher complexity levels are associated with higher headline-rate spreads.
- The right panel of Figure VI shows that products with complete-loss exposure offer higher headline returns than products without complete-loss exposure.

Interpretation: Higher headline rates are not free. They are associated with both additional payoff complexity and greater downside risk.

Economic message: The headline rate is a compensation device and a marketing device: it is increased by adding risk and complexity that are less salient than the headline yield.

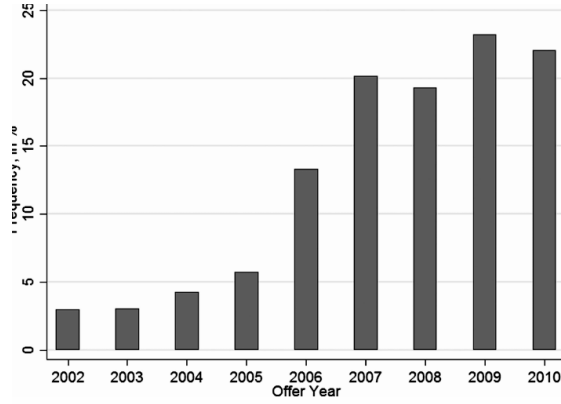


FIGURE V
Share of Volumes Exposing Investors to Total Losses

(a) Figure V: complete loss share

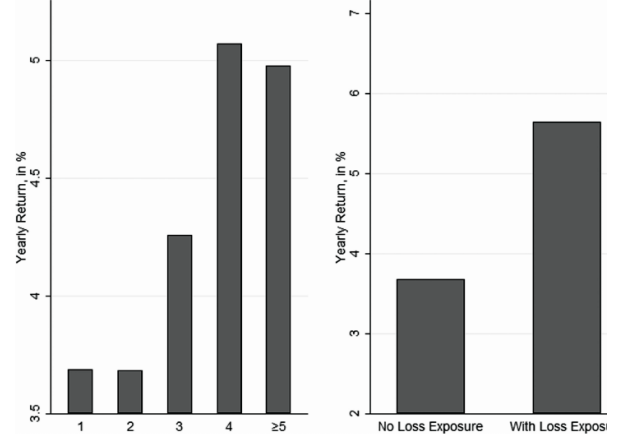


FIGURE VI
Headline Rate by Complexity Levels (Number of Features)

(b) Figure VI: headline rate by complexity

5.6 Table IV: Headline Rate, Product Complexity and Risk

Read:

- Each additional payoff feature is associated with about 0.29 percentage points of additional headline-rate spread in the controlled specification.
- Number of scenarios and description length are also positively related to headline rates.
- Complete loss exposure is associated with a much higher headline-rate spread.
- The results include product controls and fixed effects, so they are not driven only by country, year, distributor, or underlying asset composition.

Interpretation: The table formalizes the pattern in Figure VI. Complexity and risk are not just separate product characteristics; they are systematically tied to the headline return shown to investors.

Economic message: Banks can manufacture higher salient yields by adding payoff contingencies and downside exposure.

	Headline rate (spread to benchmark), in %							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Features	0.504*** (0.052)	0.291*** (0.067)						
Scenarios			0.434*** (0.058)	0.107** (0.054)				
Length (1,000 characters)					1.849*** (0.606)	0.174 (0.304)		
Complete loss exposure							1.985*** (0.160)	0.932*** (0.251)
Controls								
Distributor FE	—	Yes	—	Yes	—	Yes	—	Yes
Country FE	—	Yes	—	Yes	—	Yes	—	Yes
Underlying FE	—	Yes	—	Yes	—	Yes	—	Yes
Format FE	—	Yes	—	Yes	—	Yes	—	Yes
Maturity	—	Yes	—	Yes	—	Yes	—	Yes
Volume	—	Yes	—	Yes	—	Yes	—	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	26,352	26,352	26,352	26,352	26,352	26,352	26,352	26,352
²	0.033	0.219	0.046	0.215	0.025	0.214	0.077	0.221

Notes: This table displays the coefficients of OLS regressions in which the dependent variable is the spread between the product headline rate and the benchmark interest rate. Headline rate is defined for coupon products as the fixed yearly rate that the investor receives in the best possible scenario. The explanatory variables are the three complexity measures, as defined previously, and the measure of product risk, complete loss exposure. Standard errors are clustered at the distributor level and reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% confidence levels, respectively.

Table IV: Headline Rate, Product Complexity and Risk

5.7 Figure VII: Share of Volumes Exposing Investors to Complete Losses by Complexity Levels and Table V: Complete Loss Exposure and Product Complexity

Read:

- Figure VII shows a steep relation between complexity and complete loss exposure: higher feature counts correspond to much higher risky-volume shares.
- Table V estimates logit models and confirms that complexity predicts complete loss exposure.
- The authors remove risky features from the feature count to avoid a mechanical relation, yet the coefficients remain positive.
- Products with an additional feature have a higher probability of containing risky features even after controls.

Interpretation: The risk-complexity relation is not only a matter of counting a risk feature as complexity. Complexity more broadly accompanies downside risk.

Economic message: Complexity can make risky products more marketable by embedding risk in a conditional payoff architecture.

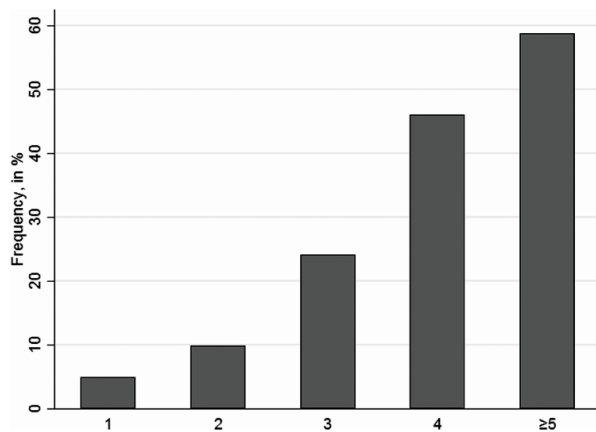


FIGURE VII

Share of Volumes Exposing Investors to Complete Losses by Complexity Levels

(a) Figure VII: complete loss by complexity

	Investor exposed to complete losses (indicator variable)					
	Logit					
	(1)	(2)	(3)	(4)	(5)	(6)
# Features	0.113* (0.067)	0.181** (0.087)				
# Scenarios			0.716*** (0.050)	0.759*** (0.068)		
Length (1,000 characters)					3.809*** (0.413)	5.876*** (0.482)
Controls						
Distributor FE	—	Yes	—	Yes	—	Yes
Country FE	—	Yes	—	Yes	—	Yes
Underlying FE	—	Yes	—	Yes	—	Yes
Maturity	—	Yes	—	Yes	—	Yes
Volume	—	Yes	—	Yes	—	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	53,541	45,163	53,541	45,519	53,541	45,519
Pseudo R^2	0.040	0.377	0.165	0.438	0.129	0.464

Notes. This table displays the coefficients of logistic regressions in which the dependent variable is a dummy equal to 1 if the product exposes the investor to complete losses. To avoid any mechanical effect, we take the conservative approach of excluding the risky features from our measure of complexity, number of features, in this analysis, and accordingly adjust downward number of scenarios. Standard errors are clustered at the distributor level and reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% confidence levels, respectively.

(b) Table V: complete loss and complexity

5.8 Table VI: Headline Rate, Complexity, and Risk in Low Interest Rate Environments

Read:

- Lower benchmark rates are associated with higher headline-rate spreads.
- A one percentage point fall in benchmark rates corresponds to roughly a 0.64 percentage point increase in headline-rate spread.
- Lower rates are also associated with more features, longer descriptions, and more complete loss exposure.

- The interaction with historical high-rate countries is negative, implying stronger effects in countries where investors were previously exposed to high yields.

Interpretation: This table is the paper’s key macro-finance mechanism. Low interest rates create a yield gap. Banks respond not by simply accepting lower advertised yields, but by designing products that preserve headline rates through complexity and risk.

Economic message: The search-for-yield environment changes the supply of securities. Retail products become more complex and riskier when safe yields decline.

TABLE VI
HEADLINE RATE, COMPLEXITY, AND RISK IN LOW INTEREST RATE ENVIRONMENTS

	Headline rate		Product complexity			Complete loss exposure	
	(Spread)		# Features	# Scenarios	Length	(Indicator)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
enchmark rate	−0.641*** (0.182)	−0.670*** (0.211)	−0.133*** (0.043)	−0.065 (0.047)	−31.900*** (9.395)	−0.506*** (0.181)	−0.655*** (0.188)
enchmark rate × historical rates		−0.026 (0.049)	−0.040*** (0.007)	−0.045*** (0.015)	−6.313*** (1.538)		−0.144** (0.064)
ontrols							
Distributor FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Underlying FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Format FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Maturity	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Volume	Yes	Yes	Yes	Yes	Yes	Yes	Yes
bservations	26,352	25,516	51,199	51,199	51,199	45,513	43,851
²	0.223	0.227	0.257	0.418	0.360	0.464	0.460

Notes: This table displays the coefficients of regressions in which the dependent variable is the spread between the product headline rate and the benchmark interest rate (five-year up rate) in the first two columns, measures of complexity in columns (3) to (5), and an indicator variable for the product exposing the investor to potential complete losses, complete exposure, in columns (6) and (7). The explanatory variable is the five-year swap rate, which takes different values in the Eurozone, the United Kingdom, Sweden, Norway, Denmark, Poland, and the Czech Republic. Historical rates is the spread between the country average interest rate prior to European monetary union and the European average. ² represents product controls and issuer, country, and year or quarter fixed effects. Standard errors are clustered at the distributor level and reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% confidence levels, respectively.

TABLE VII
COMPLEXITY MEASURES AND DISTRIBUTOR CHARACTERISTICS

	Headline rate				# Features				Complete loss exposure			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
ank type												
Savings bank	−0.171 (0.291)				0.240** (0.120)				0.407 (0.371)			
Private banking	0.305 (0.188)				0.202** (0.084)				0.497* (0.255)			
ank risk taking												
Leverage ratio	0.092** (0.039)				0.042** (0.017)				0.365*** (0.116)			
1 – deposit ratio		2.172*** (0.470)				0.923*** (0.311)			3.356*** (0.840)			
Greek exposure			4.758*** (1.201)			1.302** (0.521)			10.364*** (3.062)			
ontrols												
Underlying FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Maturity	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Volume	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
bservations	26,352	15,582	15,183	10,692	53,541	31,340	30,500	21,270	53,541	31,340	30,500	21,270
	0.117	0.114	0.125	0.141	0.093	0.119	0.121	0.113	0.167	0.239	0.220	0.282

Notes: The table displays the coefficients of OLS regressions in which the dependent variables are the three complexity measures and the explanatory variable is a dummy equal to 1 if the product distributor is a savings bank or a private banking institution (columns (1), (5) and (9))—the control group consists of commercial banks—and three proxies for risk taking: leverage, reliance on wholesale funding, and balance sheet exposure to Greek sovereign debt. Standard errors are clustered at the distributor level. *, **, and *** represent statistical significance at the 10%, 5%, and 1% confidence levels, respectively.

Table VI: Headline Rate, Complexity, and Risk in Low Interest Rate Environments

Table VII: Complexity Measures and Distributor Characteristics

5.9 Table VII: Complexity Measures and Distributor Characteristics

Read:

- Savings banks offer more complex products than commercial banks, though not significantly higher headline rates.
- Private banking institutions also distribute more complex and riskier products in some specifications.
- Bank leverage, reliance on wholesale funding, and Greek sovereign exposure are positively related to headline rates, complexity, and complete loss exposure.
- These results connect retail security design to both customer base and bank balance-sheet risk appetite.

Interpretation: Banks serving less sophisticated investors and banks with greater risk-taking indicators distribute more complex products. This supports the view that complexity is related to the seller’s incentives and the buyer’s likely sophistication.

Economic message: Complexity is not randomly distributed across the financial system. It is concentrated where investor protection concerns are especially salient.

5.10 Table VIII: Headline Rates, Complexity, and Profitability

Read:

- Higher headline-rate spreads predict higher markups.
- Each additional payoff feature raises yearly markup by about 0.34 percentage points.

- More scenarios and longer descriptions also raise markups.
- Complete loss exposure is associated with higher markups.
- In the ex post performance columns, complex products tend to deliver lower realized returns to investors.

Interpretation: This table provides the most direct evidence of the bank-profitability channel. Complexity is not just a response to investor needs; it is financially valuable to the banks distributing the products.

Economic message: Higher complexity creates rents for issuers and distributors while worsening realized outcomes for households.

TABLE VIII
HEADLINE RATES, COMPLEXITY, AND PROFITABILITY

	Markup Product yearly markup, in %					Ex post performance Product yearly return, in %				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
headline rate (spread)	0.056*					-0.051				
	(0.028)					(0.059)				
features		0.343***					-0.398***			
		(0.117)					(0.140)			
scenarios			0.196**					-0.626***		
			(0.082)					(0.212)		
length (1,000 characters)				1.256**					-2.601**	
				(0.600)					(1.181)	
complete loss exposure					0.782**					-3.496***
					(0.294)					(1.164)
controls										
loss exposure FE	—	—	—	—	—	Yes	Yes	Yes	Yes	—
distributor FE	Yes	Yes	Yes	Yes	Yes	—	—	—	—	—
credit risk	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
underlying FE	—	—	—	—	—	Yes	Yes	Yes	Yes	Yes
country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
fear FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
fear × maturity FE	—	—	—	—	—	Yes	Yes	Yes	Yes	Yes
servations	78	141	141	141	141	1,269	5,282	5,282	5,282	5,282
	0.697	0.823	0.820	0.816	0.838	0.584	0.590	0.593	0.590	0.588

Notes. This table displays the coefficients of OLS regressions in which the dependent variable is the yearly markup in columns (1) to (5) and the ex post performance in columns (6) to (10). The explanatory variables are the spread between the headline rate and the benchmark interest rate in columns (1) and (6), an indicator variable for the product exposing investor to potential total losses in columns (2) and (7), and the three complexity measures in columns (3) to (5) and (8) to (10). The sample for columns (1) to (5) consists of all debts indexed to the Euro Stoxx 50 sold in Europe in July 2009 (101 products) and a random sample of 47 products indexed to the Euro Stoxx 50 in October 2010. This sample is restricted to coupon products in column (1). Markups are computed as the difference between the offer price and product calculated fair value, obtained using the Longstaff and Schwartz OLS Monte Carlo pricing methodology (Longstaff and Schwartz 2001) with local volatility diffusion. Volatility surface data are from Eurow. The sample for columns (6) to (10) covers participation and digital products that matured before 2010. Control variables include a credit risk dummy indicating products that are noncollateralized. Standard errors are clustered at the distributor level and reported in brackets.

Table VIII: Headline Rates, Complexity, and Profitability

6 Interpretive synthesis

- **Low rates create demand pressure.** As safe yields decline, households seeking yield become more attracted to high advertised returns.
- **Banks engineer the payoff.** Derivatives allow banks to keep the headline return high while shifting probability mass into conditional downside states.
- **Complexity shrouds the trade-off.** The investor sees a simple headline number, but the risk is embedded in a conditional payoff formula.
- **Profitability reinforces the design.** Complex products have higher markups, giving banks a direct incentive to distribute them.
- **Investor sophistication matters.** Savings banks distribute more complex products, suggesting that less sophisticated clients may face more complicated designs.
- **Risk-sharing and catering can coexist.** Complexity may improve risk sharing for some investors, but the evidence suggests that catering and shrouding are central forces in this market.

7 Short summary

- The paper studies European retail structured products from 2002 to 2010.
- It develops text-based measures of headline rate, complexity, and complete loss exposure.
- Issuance volume grows rapidly over the sample.
- Headline rates remain high when benchmark interest rates fall.

- Product complexity rises over time.
- The share of products exposing investors to complete losses rises over time.
- More complex products advertise higher headline-rate spreads.
- More complex products are more likely to expose investors to complete loss.
- Low interest rates are associated with higher headline spreads, more complexity, and more risk.
- Savings banks and risk-taking banks distribute more complex and riskier products.
- Complex and risky products have higher markups.
- Complex products deliver worse ex post performance.
- The paper argues that complexity is a by-product of banks catering to yield-seeking investors.

8 Conclusion

8.1 Core conclusion

Célérier and Vallée show that retail structured products are designed around salient high headline rates and increasingly complex payoff formulas. The evidence suggests that banks use complexity and risk to maintain attractive advertised returns, especially in low-interest-rate environments.

8.2 Economic intuition

Households often pay attention to the best-case return, while the downside risk of structured products is buried in conditional payoff formulas. Banks can exploit this attention pattern by using derivative structures that make the headline rate high, while shifting downside risk into less salient states.

8.3 Incremental contribution revisited

The paper’s incremental contribution is to make the design of complex retail securities measurable at scale. It connects textual product formulas to empirical measures of headline yield, complexity, risk, markups, and performance. This allows the authors to distinguish product innovation for risk sharing from product innovation for catering and shrouding.

8.4 Policy implication

The results suggest that disclosure of a payoff formula alone may not protect households if the salient headline rate dominates attention. Consumer protection may need to focus on simplified risk disclosure, product comparability, and suitability rules for complex structured products.

8.5 Takeaway

Complexity in retail finance is not neutral. It can be a design tool that turns household demand for yield into profitable, risky, and difficult-to-understand products.